

BLESSAN CORLEY

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EDUCATION

Kalaingar Karunanidhi Institute of Technology

Coimbatore, India

B.Tech – Artificial Intelligence & Data Science — CGPA: 7.69 / 10.0

2023 – 2027

- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript

Frontend: React.js, Next.js 14, Tailwind CSS, Zustand, TanStack Query

Backend: Node.js, Express.js, REST APIs, Socket.IO, NextAuth, JWT, OAuth 2.0

Databases: PostgreSQL, MongoDB, Redis, Upstash

DevOps & Testing: Docker, GitHub Actions, Vercel, Vitest, Jest, Playwright, k6

PROJECTS

Fixly — Two-Sided Local Services Marketplace

Sep 2025 – Present | [GitHub](#)

Next.js 14 · TypeScript · MongoDB · Redis · NextAuth · Stripe · Vitest · Jest · Playwright

- Architected a write-through backend where job acceptance, payments, and notifications commit to the source of truth before fanning out to secondary systems, making downstream failures non-corrupting by design.
- Solved payment replay safety with **dual-layer idempotency** — a Redis replay key guards webhook ingestion while a unique payment record prevents duplicate entitlement grants; the idempotency key clears on unexpected failure so retries stay safe rather than getting stuck.
- Hardened auth with **live session-state refresh** — middleware re-validates account status on every request and merges live data back into the JWT, closing stale-session gaps; secured **68 routes** with **Google OAuth 2.0**, **OTP**, **bcrypt**, **CSRF protection**, and **sliding-window Redis rate limiting**.
- Offloaded async workflows to **12 background job functions** with cron jobs running within **$\pm 2s$ of schedule**; applied **fixed-window** and **sliding-window** Redis rate limiting across **60+ routes** with **Stripe webhook signature verification** and **Redis-backed payment deduplication**.
- Ensured race-condition-free hiring via an **atomic acceptance transaction** that accepts the chosen applicant, rejects all competitors, and updates usage credits in a single commit; enforced quality with **329 test suites / 1921 test cases** spanning unit, integration, and E2E layers with **zero flaky tests** in CI.

SnakrX — Game Platform with AI Bots, Leaderboards & Friend System

Oct 2025 – Feb 2026 | [GitHub](#) | [Live](#)

React 18 · Firestore / Cloud Functions · Vite · Tailwind CSS · Vitest · Playwright

- Designed a hybrid game architecture with a **requestAnimationFrame loop scaling from 200 ms/tick to 40 ms (80% latency reduction)** — browser owns real-time simulation while the server re-validates the terminal payload and finalizes scores, XP, and leaderboard writes transactionally via a deterministic session ID, eliminating duplicate saves and rank inflation.
- Eliminated all direct client writes by moving **23 Cloud Functions (22 callable + 1 scheduled)** server-side behind **deny-by-default Firestore security rules**; implemented the impossible-mode bot with **A* pathfinding** and **flood-fill BFS** for reachable-space calculation.
- Hardened signup with **hashed single-use OTP** (SHA-256, 10-min TTL, 5-attempt lockout) and per-email / IP rate limiting with full auth-user rollback on failed profile creation; engineered **idempotent stat progression** blocking replay via deterministic game IDs and applied-stats guards on both client and server.
- Gated every release on **476 passing tests** with **72% frontend** and **71% backend line coverage** — CI pipeline enforces thresholds and completes in **under 4 minutes** on every push.

NumDuel — Real-Time Multiplayer Game with Pluggable Storage

Aug 2025 – Nov 2025 | [GitHub](#) | [Live](#)

Node.js · Socket.IO · Express · PostgreSQL · Redis · Docker · GitHub Actions · Jest · Playwright

- Modeled party state as an explicit **4-phase state machine** (lobby → selection → playing → results) with **15 inbound and 34 outbound Socket.IO events** and **per-room promise locks** serializing concurrent mutations — preventing selection timeouts from overwriting submitted secrets and simultaneous round-end events from double-awarding wins.
- Designed a **pluggable storage abstraction** (MemoryStore / RedisStore) separating ephemeral session state from transactional PostgreSQL match history; implemented **EWMA latency smoothing ($\alpha=0.3$)** with signal-quality bucketing; hardened reconnection with **rotating secrets** and timing-safe hash verification, moving identity off raw socket IDs.
- Load tested with **k6** across **200 concurrent rooms** — sustained **~35ms** average event latency with **zero dropped Socket.IO events** over a 10-minute soak; Dockerized with **Compose (Node 20, Redis 7, PostgreSQL 16)**, CI completes in **under 3 minutes** gating on **424 passing tests** at **76% line coverage**.

COMPETITIVE PROGRAMMING

- **LeetCode:** Max Rating **1979** (Top 3.68%) | 450+ Problems | Best Contest Rank: **114**
- **Codeforces:** Max Rating **1630 (Expert)** | 50+ Problems | Best Contest Rank: **308** (Round 1082)
- **CodeChef:** Max Rating **2014 (5-Star)** | 250+ Problems | Best Rank: **20** | Global Rank: **1454**

ACHIEVEMENTS & CERTIFICATIONS

- **3rd Place** — REGENHACK (36-hour *solo* hackathon) at SRM Chennai among 500+ participants | **Top 50** — Code Smash DSA Hackathon among 600+ participants
- **Certifications:** Advanced DSA (Coursera) | Python Programming (NPTEL) | Advanced System Design (ByteByteGo)